Notes - Quantum Information and Quantum Fourier Transform

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May 8, 2024

Abstract

Here I collect all the interesting stuff regarding the non-stabilizerness of the quantum Fourier Transform.

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1 Useful definitions

- **Q projector**

$$Q = \frac{1}{d^2} \sum_{P \in \mathcal{P}_n} P^{\otimes 4}, \tag{1.1}$$

where $\mathcal{P}_n$ is the quotient group (strings with not phases, i.e. ± 1 and $\pm i$) of the n-qubit Pauli group.

- T_π **permutation operator** The permutation operators $\mathrm{T}_\pi \in \mathcal{B}\left(\mathcal{H}^{\otimes k}\right)$ are built in a similar fashion as the swap operator acting on $\mathcal{H}^{\otimes 2}$: given a permutation $\pi \in S_k$,

$$\mathrm{T}_\pi \left|i_1, \ldots, i_k\right\rangle = \left|\pi\left(i_1\right), \ldots, \pi\left(i_k\right)\right\rangle .$$

Now, define a state $\psi \in \mathcal{H}$ as $\psi = \sum_i \psi_i |i\rangle\langle i|$: its purity is naturally equal to $\sum_i \psi_i^2$. Its counterpart in $\mathcal{H}^{\otimes 2}$ reads

$$\psi^{\otimes 2} = \sum_{ij} \psi_i \psi_j |ij\rangle\langle ij|.$$

Calculation the expectation value of the swap operator on $\psi^{\otimes 2}$ yields

$$\mathrm{Tr}\, \mathrm{T}\, \psi^{\otimes 2} = \sum_{ij} \psi_i \psi_j \,\mathrm{Tr}\, |ji\rangle\langle ij| = \sum_{ij} \psi_i \psi_j \delta_{ij} = \sum_i \psi_i^2 = \mathrm{Pur}\, \psi.$$

- Π_{sym} **symmetric projector**

$$\Pi_{\text{sym}} = \frac{1}{4!} \sum_{\pi \in S_4} T_\pi \tag{1.2}$$

is the projector on the symmetric subspaces of $\mathcal{H}^{\otimes 4}$ under the permutation of 4 objects.

2 Fourier Transform - TLDR

2.1 Introduction

Discrete Fourier Transform takes an input a vector of complex numbers of length N and outputs a vector of complex numbers after applying a transformation as

$$y_k := \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{\frac{2\pi i j k}{N}}. \tag{2.1}$$

For example, lets take the input vector with $N = 4$ elements

$$\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.2}$$

The Fourier transformation yields

$$y_{k=0} = \frac{1}{\sqrt{4}} \sum_{j=0}^{3} x_j e^{\frac{2\pi i j}{2} 0} = \frac{1}{2} \sum_{j=0}^{3} x_j = \frac{1}{2} \tag{2.3}$$

$$y_{k=1} = \frac{1}{\sqrt{2}} \sum_{j=0}^{3} x_j e^{\frac{2\pi i j}{2} 1} = \frac{1}{2} \left(x_0 + x_1 \cancelto{0}{e^{i\pi}} + x_2 \cancelto{0}{e^{2i\pi}} + x_2 \cancelto{0}{e^{3i\pi}} \right) = \frac{1}{2} \tag{2.4}$$

$$y_{k=2} = \frac{1}{\sqrt{2}} \sum_{j=0}^{3} x_j e^{\frac{2\pi i j}{2} 2} = \frac{1}{2} \left(x_0 + x_1 \cancelto{0}{e^{2i\pi}} + x_2 \cancelto{0}{e^{4i\pi}} + x_2 \cancelto{0}{e^{6i\pi}} \right) = \frac{1}{2} \tag{2.5}$$

$$y_{k=3} = \frac{1}{\sqrt{2}} \sum_{j=0}^{3} x_j e^{\frac{2\pi i j}{2} 3} = \frac{1}{2} \left(x_0 + x_1 \cancelto{0}{e^{3i\pi}} + x_2 \cancelto{0}{e^{6i\pi}} + x_2 \cancelto{0}{e^{9i\pi}} \right) = \frac{1}{2} \tag{2.6}$$

$$\tag{2.7}$$

therefore we have that Fourier Transform transforms the vector as

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \longrightarrow \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}. \tag{2.8}$$

The Quantum Fourier Transform is the same operation but acting on an orthonormal basis as

$$|j\rangle = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} e^{\frac{2\pi i j k}{d}} |k\rangle, \tag{2.9}$$

where $d = 2^N$ for N qubits.

2.2 Quantum Fourier transform unitary

In general we can write the unitary (see Ref. [5]) as

$$U^{\mathrm{FT}}(d) = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \sum_{j=0}^{d-1} \omega^{jk}(d) |k\rangle\langle j|, \tag{2.10}$$

where $\omega^{jk}(d) = e^{\frac{2\pi i}{d} jk}$ is the root-of-unity multiplicative scalar constants or *twiddle* factor. For example for a single qubit $N = 1$ the unitary reads

$$U^{\mathrm{FT}}(2^1) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{2.11}$$

For two qubits ($N = 2$) reads

$$U^{\mathrm{FT}}(2^2) = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}. \tag{2.12}$$

While for $N = 3$ it reads

$$U^{\mathrm{FT}}(2^3) = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \omega^4 & \omega^5 & \omega^6 & \omega^7 \\ 1 & \omega^2 & \omega^4 & \omega^6 & 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega & \omega^4 & \omega^7 & \omega^2 & \omega^5 \\ 1 & \omega^4 & 1 & \omega^4 & 1 & \omega^4 & 1 & \omega^4 \\ 1 & \omega^5 & \omega^2 & \omega^7 & \omega^4 & \omega & \omega^6 & \omega^3 \\ 1 & \omega^6 & \omega^4 & \omega^2 & 1 & \omega^6 & \omega^4 & \omega^2 \\ 1 & \omega^7 & \omega^6 & \omega^5 & \omega^4 & \omega^3 & \omega^2 & \omega \end{bmatrix}. \tag{2.13}$$

2.2.1 Circuit structure and approximation

To implement a n-qubit Fourier transform we require the application of a polynomial number of gates $\mathcal{O}(n^2)$ quantum gates (see Fig. 1 or page 219 in Ref. [3]). The quantum Fourier transform consists of a nested circuit layers that for each qubit requires a gate of a smaller rotation $R_2, R_3, R_4, R_5, ...$ This allows for a controlled and coherent approximation by neglecting gates of with small phases inducing an error ε. Furthermore, not all gates are 'born' equal and implementation of Clifford gates are considered cheap, while T gates expensive. From this point of view we can approximate in the phase angle (up to the approximation error ε) the quantum Fourier transform with the number of T gates denoted as $t(U)$ (see Ref. [4]) as

$$t(U^{\mathrm{FT}}(n)) \sim \mathcal{O}(n \log(n)), \tag{2.14}$$

where we suppressed the approximation error ε.

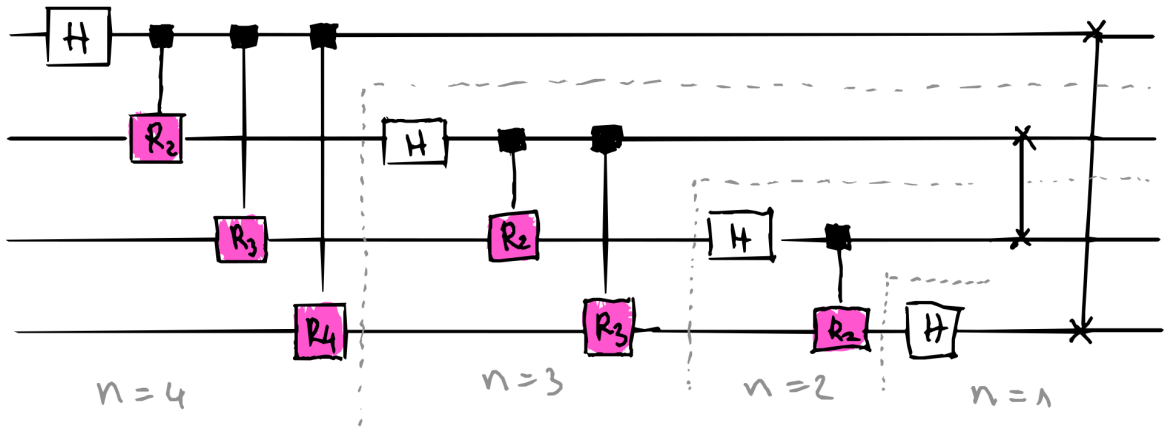


Figure 1: Quantum Fourier Transform circuit of four qubits $U^{\mathrm{FT}}(4)$. We see that the FT has an embedded structure of smaller circuits $n = 1, 2, 3$ present in the $n = 4$ version. With purple we denote the magic inducing gates (see next sections). This can be shown by computing the commutator $\left[U^{\otimes 4}, Q\Pi_{\mathrm{sym}}\right]$ and showing its non-vanishing for each layer. The rotation gates are defined as $R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$. Additionally, we identify $R_2 = S$, $R_3 = T$. Here H denotes the Hadamard gate and crosses connecting qubits are SWAP gates. The number of gates is $n + (n-1) + (n-2) + ... + 1 = \frac{n(n+1)}{2}$, plus the SWAP gates that number $n/2$. This implies that leading order scaling of the total number of gates in the circuit scales with number of qubits as $\mathcal{O}(n^2)$.

3 Non-stablizing power of the Fourier Transform (FT)

3.1 1-qubit FT

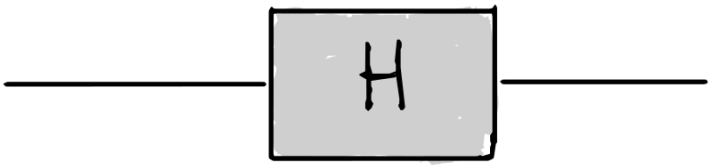


Figure 2: The quantum Fourier transform for one qubit $U^{\mathrm{FT}}(1)$ in this case is just a Hadamard gate acting on the computational basis states (eigenstates of Pauli σ^z operator $\{0,1\}$). From Ref. [3].

In particular, Eq. (2.9) reads for $N = 1$

$$|j\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^{1} e^{ij\pi k} |k\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{ij\pi} |1\rangle \right). \tag{3.1}$$

The computational basis states can be either $j \in \{0,1\}$ so FT behaves like a Hadamard gate

$$H|0\rangle = |+\rangle, \tag{3.2}$$

$$H|1\rangle = |-\rangle. \tag{3.3}$$

Hadamard gate is defined

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{X+Z}{\sqrt{2}}. \tag{3.4}$$

3.1.1 Non-stabilizing power calculation

From Ref. [1] we know that linearized Stabilizer Rényi entropy (and by extension the SRE itself) is vanishing when

$$M_{\mathrm{lin}}(U) = 0 \quad \text{iff} \quad [U^{\otimes 4}, Q\Pi_{\mathrm{sym}}] = 0 \tag{3.5}$$

for a given unitary U. For the case of the 1 qubit FT transform this unitary is the Hadamard gate. To show the that the stabilizing power of the 1 qubit FT is zero we need to prove

$$\left[\left(U^{\mathrm{FT}}(1) \right)^{\otimes 4}, Q\Pi_{\mathrm{sym}} \right] = \left[H^{\otimes 4}, Q\Pi_{\mathrm{sym}} \right] \stackrel{!}{=} 0. \tag{3.6}$$

We first specify things for this case

- projection operator

$$Q = \frac{1}{4} \sum_{P \in \mathcal{P}} P^{\otimes 4} = \frac{1}{4} \left(I^{\otimes 4} + X^{\otimes 4} + Y^{\otimes 4} + Z^{\otimes 4} \right) \tag{3.7}$$

- permutation operator

$$T_\pi = \sum_{i,j,k,l} |\pi(ijkl)\rangle\langle ijkl| \tag{3.8}$$

We proceed as

$$\begin{aligned}\left[H^{\otimes 4}, Q\Pi_{\text{sym}}\right] &= \left[H^{\otimes 4}, \frac{1}{4}\sum_{P\in\mathcal{P}} P^{\otimes 4}\frac{1}{4!}\sum_{\pi\in S_4} T_\pi\right] = \frac{1}{96}\sum_{P\in\mathcal{P}}\sum_{\pi\in S_4}\left[H^{\otimes 4}, P^{\otimes 4}T_\pi\right] \\ &= \frac{1}{96}\sum_{P\in\mathcal{P}}\sum_{\pi\in S_4}\left(\left[H^{\otimes 4}, P^{\otimes 4}\right]T_\pi + P^{\otimes 4}\left[H^{\otimes 4}, T_\pi\right]\right).\end{aligned} \tag{3.9}$$

As the second terms gives us zero we have

$$\left[H^{\otimes 4}, Q\Pi_{\text{sym}}\right] = \frac{1}{96}\sum_{P\in\mathcal{P}}\sum_{\pi\in S_4}\left[H^{\otimes 4}, P^{\otimes 4}\right]T_\pi. \tag{3.10}$$

NOTE 1:
The second term reads

$$\left[H^{\otimes 4}, T_\pi\right] = H^{\otimes 4}T_\pi - T_\pi H^{\otimes 4}, \tag{3.11}$$

implying

$$H^{\otimes 4}T_\pi = T_\pi H^{\otimes 4}. \tag{3.12}$$

Multiplying with the identity $\mathbb{I} = T_\pi^\dagger T_\pi$ from the left

$$H^{\otimes 4}T_\pi = \underbrace{T_\pi H^{\otimes 4}T_\pi^\dagger}_{H^{\otimes 4}}T_\pi, \quad \longrightarrow \left[H^{\otimes 4}, T_\pi\right] = 0 \tag{3.13}$$

NOTE 2: For the first term we have to focus on the commutator that we can write as

$$\left[H^{\otimes 4}, P^{\otimes 4}\right] = \left[H^{\otimes 2} \otimes H^{\otimes 2}, P^{\otimes 2} \otimes P^{\otimes 2}\right], \tag{3.14}$$

a form that allows us to take advantage of the identities from Appendix from Ref. [2]

$$[A_1 \otimes A_2, B_1 \otimes B_2] = \frac{1}{2}\left([A_1, B_1] \otimes \{A_2, B_2\} + \{A_1, B_1\} \otimes [A_2, B_2]\right), \tag{3.15}$$

$$\{A_1 \otimes A_2, B_1 \otimes B_2\} = \frac{1}{2}\left([A_1, B_1] \otimes [A_2, B_2] + \{A_1, B_1\} \otimes \{A_2, B_2\}\right). \tag{3.16}$$

Using the first identity twice we obtain

$$\begin{aligned}\left[H^{\otimes 4}, P^{\otimes 4}\right] &= \frac{1}{2}([H^{\otimes 2}, P^{\otimes 2}] \otimes \{H^{\otimes 2}, P^{\otimes 2}\} + \{H^{\otimes 2}, P^{\otimes 2}\} \otimes [H^{\otimes 2}, P^{\otimes 2}]) \\ &= \frac{1}{4}\Big[\Big([H,P] \otimes \{H,P\} + \{H,P\} \otimes [H,P]\Big) \otimes \{H^{\otimes 2}, P^{\otimes 2}\} \\ &+ \{H^{\otimes 2}, P^{\otimes 2}\} \otimes \Big([H,P] \otimes \{H,P\} + \{H,P\} \otimes [H,P]\Big)\Big].\end{aligned} \tag{3.17}$$

Now apply the second identity for the anti-commutator

$$\begin{aligned}\left[H^{\otimes 4}, P^{\otimes 4}\right] = \frac{1}{8}\Big(&[H,P] \otimes \{H,P\} \otimes [H,P] \otimes [H,P] \\ &+ [H,P] \otimes \{H,P\} \otimes \{H,P\} \otimes \{H,P\} \\ &+ \{H,P\} \otimes [H,P] \otimes [H,P] \otimes [H,P] \\ &+ \{H,P\} \otimes [H,P] \otimes \{H,P\} \otimes \{H,P\} \\ &+ [H,P] \otimes [H,P] \otimes [H,P] \otimes \{H,P\} \\ &+ [H,P] \otimes [H,P] \otimes \{H,P\} \otimes [H,P] \\ &+ \{H,P\} \otimes \{H,P\} \otimes [H,P] \otimes \{H,P\} \\ &+ \{H,P\} \otimes \{H,P\} \otimes \{H,P\} \otimes [H,P]\Big)\end{aligned} \tag{3.18}$$

We have therefore

$$\begin{aligned}\left[H^{\otimes 4}, Q\Pi_{\text{sym}}\right] = \frac{1}{768}\sum_{P\in\mathcal{P}}\sum_{\pi\in S_4}\Big(&[H,P] \otimes \{H,P\} \otimes [H,P] \otimes [H,P] \\ &+ [H,P] \otimes \{H,P\} \otimes \{H,P\} \otimes \{H,P\} \\ &+ \{H,P\} \otimes [H,P] \otimes [H,P] \otimes [H,P] \\ &+ \{H,P\} \otimes [H,P] \otimes \{H,P\} \otimes \{H,P\} \\ &+ [H,P] \otimes [H,P] \otimes [H,P] \otimes \{H,P\} \\ &+ [H,P] \otimes [H,P] \otimes \{H,P\} \otimes [H,P] \\ &+ \{H,P\} \otimes \{H,P\} \otimes [H,P] \otimes \{H,P\} \\ &+ \{H,P\} \otimes \{H,P\} \otimes \{H,P\} \otimes [H,P]\Big) T_\pi.\end{aligned} \tag{3.19}$$

NOTE 3: Lets focus on the term

$$\sum_P ([H,P]\otimes\{H,P\}\otimes\{H,P\}\otimes\{H,P\}) \tag{3.20}$$

when written out reads

$$\begin{aligned}
&= ([H,I]\otimes\{H,I\}\otimes\{H,I\}\otimes\{H,I\})\\
&+ ([H,X]\otimes\{H,X\}\otimes\{H,X\}\otimes\{H,X\})\\
&+ ([H,Y]\otimes\{H,Y\}\otimes\{H,Y\}\otimes\{H,Y\})\\
&+ ([H,Z]\otimes\{H,Z\}\otimes\{H,Z\}\otimes\{H,Z\})\,.
\end{aligned}$$

Using commutator identities

$$\begin{aligned}
[H,X] &= i\sqrt{2}Y,\\
[H,Y] &= i\sqrt{2}(Z-X),\\
[H,Z] &= -i\sqrt{2}Y,
\end{aligned}$$

and anti-commutation identities

$$\begin{aligned}
\{H,X\} &= 2I\\
\{H,Y\} &= 0\\
\{H,Z\} &= 2I
\end{aligned}$$

Hence we obtain for the term in Eq. (3.20)

$$\begin{aligned}
\sum_P ([H,P]\otimes\{H,P\}\otimes\{H,P\}\otimes\{H,P\}) &= 0\otimes 0\otimes 0\otimes 0 + 8(i\sqrt{2})\,(Y\otimes I\otimes I\otimes I)\\
&+ (i\sqrt{2})\,((Z-X)\otimes 0\otimes 0\otimes 0) - 8(i\sqrt{2})\,(Y\otimes I\otimes I\otimes I)\\
&= 0
\end{aligned} \tag{3.21}$$

The remaining 7 terms follow similarly and yield a zero.

From this calculation we see that indeed

$$\boxed{\left[\left(U^{\mathrm{FT}}(1)\right)^{\otimes 4}, Q\Pi_{\mathrm{sym}}\right] = 0,} \tag{3.22}$$

implying that non-stabilizing power $M_{\mathrm{lin}}(U^{\mathrm{FT}}(1)) = 0$ of a $N = 1$ qubit Fourier transform is zero. This is consistent with the fact that the Hadamard is a Clifford unitary.

3.2 2-qubit FT

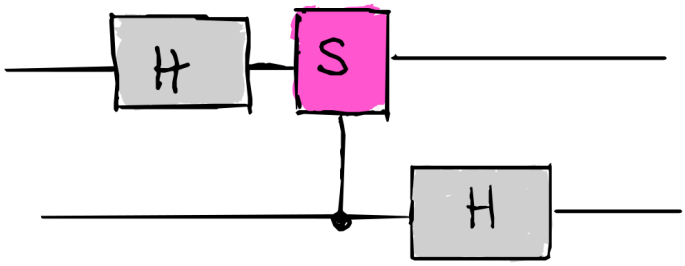


Figure 3: Quantum Fourier Transform circuit of two qubits $U^{\mathrm{FT}}(2)$. With purple we denote the magic inducing gate (see below). More info in Ref. [3]. Note there is a difference (reversal) in the definition of the control+s as compared to the Fig. 1.

From $N \geq 2$ qubits it becomes convenient to express the FT using the notation from Ref. [3] as

$$|j\rangle = \frac{1}{2^{N/2}} \left(|0\rangle + e^{2\pi i 0.j_N}|1\rangle\right) \otimes \left(|0\rangle + e^{2\pi i 0.j_{N-1}j_N}|1\rangle\right) \otimes ... \otimes \left(|0\rangle + e^{2\pi i 0.j_1 j_2 ... j_N}|1\rangle\right), \tag{3.23}$$

where the notation implies representation of the binary fraction as

$$0.j_l j_{l+1} ... j_m := \frac{j_l}{2} + \frac{j_{l+1}}{4} + ... + \frac{j_m}{2^{m-l+1}}. \tag{3.24}$$

Note that here $j_l \in \{0, 1\}$ and other are binary numbers. More specifically lets take $N = 2$ case and the state $|j_1 j_2\rangle$. Here we are using binary representation where $|j\rangle = |j_1 j_2\rangle$. For example

$$\begin{aligned} |j = 0\rangle &\longrightarrow |00\rangle \\ |j = 1\rangle &\longrightarrow |01\rangle \\ |j = 2\rangle &\longrightarrow |10\rangle \\ |j = 3\rangle &\longrightarrow |11\rangle \end{aligned}$$

where the transition between the index to binary representation is set by the function

$$j = j_1 2^{N-1} + j_2 2^{N-2} + ... + j_N 2^0. \tag{3.25}$$

Now we can build the FT circuit from the $|j_1 j_2\rangle$ input by first applying a Hadamard gate to the first qubit and we obtain

$$\frac{1}{2^{1/2}} \left(|0\rangle + e^{2\pi i 0.j_1}|1\rangle\right) |j_2\rangle. \tag{3.26}$$

Next we apply a $R - k$ gate of the form

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^k}} \end{bmatrix}, \tag{3.27}$$

where we fix $k = 2$ this is the the phase gate

$$R_2 = S = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{\pi i}{2}} \end{bmatrix} \tag{3.28}$$

$R_2 = S$. However, this gate is applied while keeping in mind the value of the first second qubit so we need to apply control-S. Afterwards, one needs to apply Hadamard on the second qubit. We obtain

$$|j\rangle = |j_1 j_2\rangle = \frac{1}{2}\left(|0\rangle + e^{2\pi i 0.j_2}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.j_1 j_2}|1\rangle\right). \tag{3.29}$$

The generating circuit is given in Fig. 3.

3.2.1 Non-stabilizing power calculation

We check if

$$\left[\left(U^{\mathrm{FT}}(2)\right)^{\otimes 4}, Q\Pi_{\mathrm{sym}}\right] = \frac{1}{16}\frac{1}{4!}\sum_{\pi\in S_4}\sum_{P\in\mathcal{P}(2)}\left[\left(U^{\mathrm{FT}}(2)\right)^{\otimes 4}, P^{\otimes 4}\right]T_\pi \stackrel{!}{=} 0, \tag{3.30}$$

Where we have use the fact that the the unitary is not affected by the permutation operator (see Note 1). The projection operator is defined as

$$\begin{aligned} Q = \frac{1}{16}\sum_{P\in\mathcal{P}(2)} P^{\otimes 4} = &\frac{1}{16}\Big((II)^{\otimes 4} + (IX)^{\otimes 4} + (IY)^{\otimes 4} + (IZ)^{\otimes 4} \\ &+ (XI)^{\otimes 4} + (XX)^{\otimes 4} + (XY)^{\otimes 4} + (XZ)^{\otimes 4} \\ &+ (YI)^{\otimes 4} + (YX)^{\otimes 4} + (YY)^{\otimes 4} + (YZ)^{\otimes 4} \\ &+ (ZI)^{\otimes 4} + (ZX)^{\otimes 4} + (ZY)^{\otimes 4} + (ZZ)^{\otimes 4}\Big). \end{aligned} \tag{3.31}$$

Using the commutation relations (see Note 2) we obtain

$$\begin{aligned} \left[\left(U^{\mathrm{FT}}(2)\right)^{\otimes 4}, Q\Pi_{\mathrm{sym}}\right] = \frac{1}{16}\frac{1}{4!}\sum_{\pi\in S_4}\sum_{P\in\mathcal{P}(2)}\left[\left(U^{\mathrm{FT}}(2)\right)^{\otimes 4}, P^{\otimes 4}\right]T_\pi = \frac{1}{16}\frac{1}{4!}\frac{1}{8}\sum_{\pi\in S_4}\sum_{P\in\mathcal{P}(2)}\Big(\\ [U^{\mathrm{FT}}(2),P]\otimes\{U^{\mathrm{FT}}(2),P\}\otimes[U^{\mathrm{FT}}(2),P]\otimes[U^{\mathrm{FT}}(2),P] \\ + [U^{\mathrm{FT}}(2),P]\otimes\{U^{\mathrm{FT}}(2),P\}\otimes\{U^{\mathrm{FT}}(2),P\}\otimes\{U^{\mathrm{FT}}(2),P\} \\ + \{U^{\mathrm{FT}}(2),P\}\otimes[U^{\mathrm{FT}}(2),P]\otimes[U^{\mathrm{FT}}(2),P]\otimes[U^{\mathrm{FT}}(2),P] \\ + \{U^{\mathrm{FT}}(2),P\}\otimes[U^{\mathrm{FT}}(2),P]\otimes\{U^{\mathrm{FT}}(2),P\}\otimes\{U^{\mathrm{FT}}(2),P\} \\ + [U^{\mathrm{FT}}(2),P]\otimes[U^{\mathrm{FT}}(2),P]\otimes[U^{\mathrm{FT}}(2),P]\otimes\{U^{\mathrm{FT}}(2),P\} \\ + [U^{\mathrm{FT}}(2),P]\otimes[U^{\mathrm{FT}}(2),P]\otimes\{U^{\mathrm{FT}}(2),P\}\otimes[U^{\mathrm{FT}}(2),P] \\ + \{U^{\mathrm{FT}}(2),P\}\otimes\{U^{\mathrm{FT}}(2),P\}\otimes[U^{\mathrm{FT}}(2),P]\otimes\{U^{\mathrm{FT}}(2),P\} \\ + \{U^{\mathrm{FT}}(2),P\}\otimes\{U^{\mathrm{FT}}(2),P\}\otimes\{U^{\mathrm{FT}}(2),P\}\otimes[U^{\mathrm{FT}}(2),P]\Big)T_\pi \stackrel{!}{=} 0. \end{aligned} \tag{3.32}$$

The FT transformation takes the following form

$$U^{\mathrm{FT}}(2) = \underbrace{(I\otimes H)}_{\text{1.layer}}\underbrace{\Big(|0\rangle\langle 0|\otimes S + |1\rangle\langle 1|\otimes I\Big)}_{\text{2.layer}}\underbrace{(H\otimes I)}_{\text{3.layer}}. \tag{3.33}$$

where $|k_1 k_2\rangle = U^{\mathrm{FT}}(2)|j_1 j_2\rangle = U_1 U_2 U_3 |j_1 j_2\rangle$. Using expressions from Appendix A we have

$$\{U_1U_2U_3, P\} = U_1U_2\{U_3,P\} - U_1[U_2,P]U_3 - [U_1,P]U_2U_3, \tag{3.34}$$

$$[U_1U_2U_3, P] = U_1U_2[U_3,P] + U_1[U_2,P]U_3 + [U_1,P]U_2U_3. \tag{3.35}$$

At this point we can use the fact that non-stabilizing power exhibits stability under Clifford operations, i.e. $M(UC) = M(U)$. In Appendix B, we show that layer 1 circuit operator U_1 has zero a commutator just for fun (follows the same calculation as for the 1 qubit case), but from the definition of the Hadamard in Eq. (3.4) we know it is a Clifford unitary.
Naturally focus on the second layer

$$\begin{aligned}\left[(U_2)^{\otimes 4}, Q\Pi_{\text{sym}}\right] = \frac{1}{16}\frac{1}{4!}\frac{1}{8}\sum_{P\in\mathcal{P}(2)}\sum_{\pi\in S_4}\left[(U_2)^{\otimes 4}, P^{\otimes 4}\right]T_\pi = \frac{1}{16}\frac{1}{4!}\frac{1}{8}\sum_{\pi\in S_4}\sum_{P\in\mathcal{P}(2)}\Big(\\ [U_2,P]\otimes\{U_2,P\}\otimes[U_2,P]\otimes[U_2,P]\\ +[U_2,P]\otimes\{U_2,P\}\otimes\{U_2,P\}\otimes\{U_2,P\}\\ +\{U_2,P\}\otimes[U_2,P]\otimes[U_2,P]\otimes[U_2,P]\\ +\{U_2,P\}\otimes[U_2,P]\otimes\{U_2,P\}\otimes\{U_2,P\}\\ +[U_2,P]\otimes[U_2,P]\otimes[U_2,P]\otimes\{U_2,P\}\\ +[U_2,P]\otimes[U_2,P]\otimes\{U_2,P\}\otimes[U_2,P]\\ +\{U_2,P\}\otimes\{U_2,P\}\otimes[U_2,P]\otimes\{U_2,P\}\\ +\{U_2,P\}\otimes\{U_2,P\}\otimes\{U_2,P\}\otimes[U_2,P]\Big)T_\pi\end{aligned} \tag{3.36}$$

NOTE 5: Writing the appearing commutator we have

$$\Big[U_2, P\Big] = \Big[(|0\rangle\langle 0| \otimes S + |1\rangle\langle 1| \otimes I), A \otimes B\Big] = \Big[|0\rangle\langle 0| \otimes S, A \otimes B\Big] + \Big[|1\rangle\langle 1| \otimes I, A \otimes B\Big]$$
$$= \frac{1}{2}\Big(\Big[|0\rangle\langle 0|, A\Big] \otimes \{S, B\} + \{|0\rangle\langle 0|, A\} \otimes \Big[S, B\Big] + \Big[|1\rangle\langle 1|, A\Big] \otimes \{I, B\}\Big). \tag{3.37}$$

The anti-commutator on the other hand read

$$\Big\{U_2, P\Big\} = \Big\{(|0\rangle\langle 0| \otimes S + |1\rangle\langle 1| \otimes I), P\Big\} = \Big\{|0\rangle\langle 0| \otimes S, A \otimes B\Big\} + \Big\{|1\rangle\langle 1| \otimes I, A \otimes B\Big\}$$
$$= \frac{1}{2}\Big(\Big[|0\rangle\langle 0|, A\Big] \otimes [S, B] + \Big\{|0\rangle\langle 0|, A\Big\} \otimes \{S, B\} + \Big\{|1\rangle\langle 1|, A\Big\} \otimes \{I, B\}\Big) \tag{3.38}$$

Using this decompositions we see each of the 8 terms in the Eq. (3.36) decomposes in the 81 term. In total that means we have $8 \times 81 = 648$ terms to handle.

Next we list the possibilities for the basic commutators appearing in this monster of a relation

$$\begin{aligned}
\Big[|0\rangle\langle 0|, I\Big] &= 0, \qquad \Big\{|0\rangle\langle 0|, I\Big\} = I + Z \\
\Big[|0\rangle\langle 0|, X\Big] &= iY, \qquad \Big\{|0\rangle\langle 0|, X\Big\} = X \\
\Big[|0\rangle\langle 0|, Y\Big] &= -iX, \qquad \Big\{|0\rangle\langle 0|, Y\Big\} = Y \\
\Big[|0\rangle\langle 0|, Z\Big] &= 0, \qquad \Big\{|0\rangle\langle 0|, Z\Big\} = I + Z
\end{aligned} \tag{3.39}$$

The next ones are

$$\begin{aligned}
\Big[|1\rangle\langle 1|, I\Big] &= 0, \qquad \Big\{|1\rangle\langle 1|, I\Big\} = I - Z \\
\Big[|1\rangle\langle 1|, X\Big] &= -iY, \qquad \Big\{|1\rangle\langle 1|, X\Big\} = X \\
\Big[|1\rangle\langle 1|, Y\Big] &= iX, \qquad \Big\{|1\rangle\langle 1|, Y\Big\} = Y \\
\Big[|1\rangle\langle 1|, Z\Big] &= 0, \qquad \Big\{|1\rangle\langle 1|, Z\Big\} = -(I - Z)
\end{aligned} \tag{3.40}$$

The final with the phase gate read

$$\begin{aligned}
\Big[S, I\Big] &= 0, \qquad \Big\{S, I\Big\} = I + Z + i(I - Z) \\
\Big[S, X\Big] &= (i+1)Y, \qquad \Big\{S, X\Big\} = (1+i)X \\
\Big[S, Y\Big] &= -(i+1)X, \qquad \Big\{S, Y\Big\} = (1+i)Y \\
\Big[S, Z\Big] &= 0, \qquad \Big\{S, Z\Big\} = I + Z - i(I - Z)
\end{aligned} \tag{3.41}$$

Using this information we can build all the terms in Eq. (3.36) using this basic commutators. Using Mathematica we can handle such a larger number of terms analytically.

Following the strategy in Note 5 we obtain

$$\left[(U_2)^{\otimes 4}, Q\Pi_{\text{sym}}\right] \neq 0, \tag{3.42}$$

and is non-zero matrix as can be seen in Fig. 4. Explicit form can be found in Mathematica notebook. We have verified that the analytical form is correct by computing the commutator numerically. If we now compute the Frobenious norm $\|A\|_2 = \sqrt{\text{Tr}\, A^\dagger A}$ of this commutator, i.e.

$$\left|\left|\left[(U_2)^{\otimes 4}, Q\Pi_{\text{sym}}\right]\right|\right|_2^2 = 6. \tag{3.43}$$

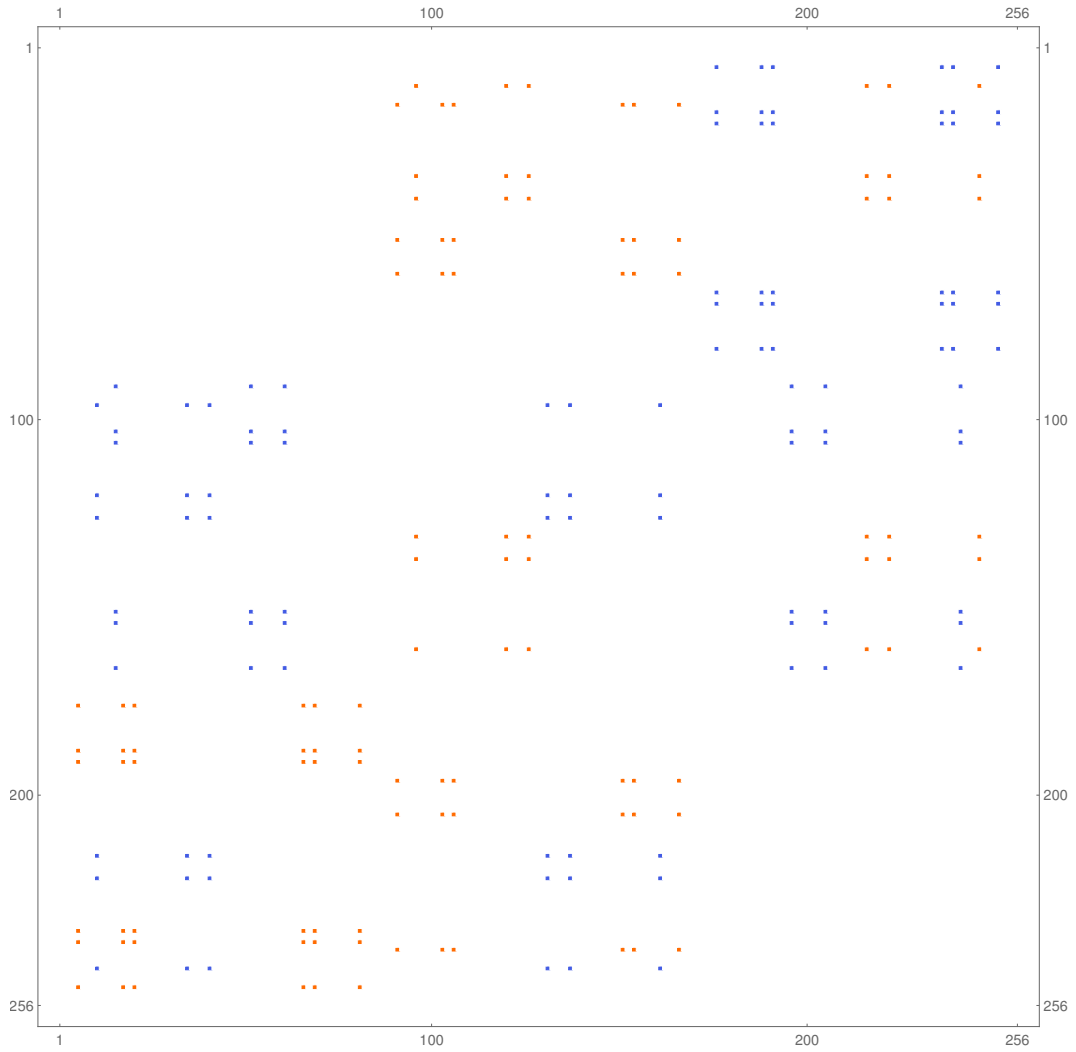


Figure 4: Commutator $\left[(U_2)^{\otimes 4}, Q\Pi_{\text{sym}}\right]$ we obtain. There are two types of values appearing $\pm\frac{1}{6}$. There is an equal number of positive and negative contributions.

3.2.2 Linear non-stabilizing power

According to the definition in Ref. [1] (see Eq. (S.39)) we have

$$\mathcal{M}_{\text{lin}}\left(U\right)=\frac{d}{2(4+d)\,\text{tr}\left(Q\Pi_{\text{sym}}\right)}\left\|\left[U^{\otimes 4},Q\Pi_{\text{sym}}\right]\right\|_2^2, \tag{3.44}$$

where $\text{tr}\left(Q\Pi_{\text{sym}}\right)=\frac{(d+1)(d+2)}{6}$. Using the the result from Eq.(3.43) we obtain for the second layer of the FT the following value

$$\mathcal{M}_{\text{lin}}\left(U_2\right)=\frac{18d}{(d+1)(d+2)(4+d)} \tag{3.45}$$

We know that CONTROL+S (always a 2 qubit unitary $d=2^2$) element in N qubit FT always exists (see 3 and 4 qubit circuits in Figs. 5 and 6) so the minimal amount of linearized non-stabilizing power induced by quantum FT unitary is

$$\min\mathcal{M}_{\text{lin}}\left(U^{\text{FT}}(N)\right)=\frac{3}{10}. \tag{3.46}$$

Which is less than a typical 2-qubit (Haar) unitary (see Eqs (S.48)) that yield $\mathbb{E}_U\left[\mathcal{M}_{\text{lin}}(U)\right]=\frac{3}{7}$, but more than a typical 1-qubit gate with $\mathbb{E}_U\left[\mathcal{M}_{\text{lin}}(U)\right]=1/5$. Note that with the increase of the quantum FT circuit more gates come into the game inducing more non-stabilizerness and therefore Eq. (3.46) provides a minimal amount of linear non-stabilizing power of the quantum Fourier transform.

3.3 3-qubit FT

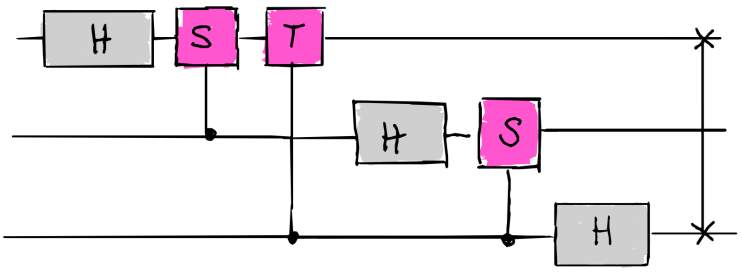


Figure 5: Quantum Fourier Transform circuit of three qubits $U^{\mathrm{FT}}(3)$. More info in Ref. [3]. With purple we denote the magic inducing gate. Last gate applied is a swap gate between the first and the third qubit so have correctly qubits. Note there is a difference (reversal) in the definition of the control+rotation as compared to the Fig. 1.

3.4 4-qubit FT

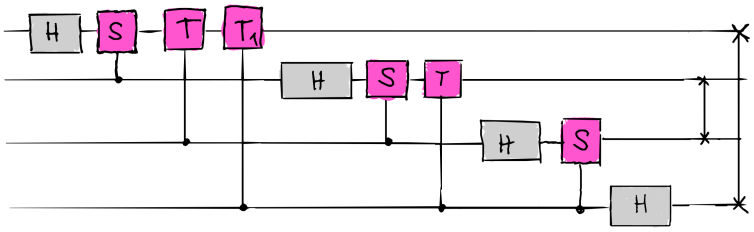


Figure 6: Quantum Fourier Transform circuit of four qubits $U^{\mathrm{FT}}(4)$. More info in Ref. [3]. With purple we denote the magic inducing T gate. Another magic inducing gate is T_1 which has a phase factor of half, of that of a T gate. Note there is a difference (reversal) in the definition of the control+rotation as compared to the Fig. 1.

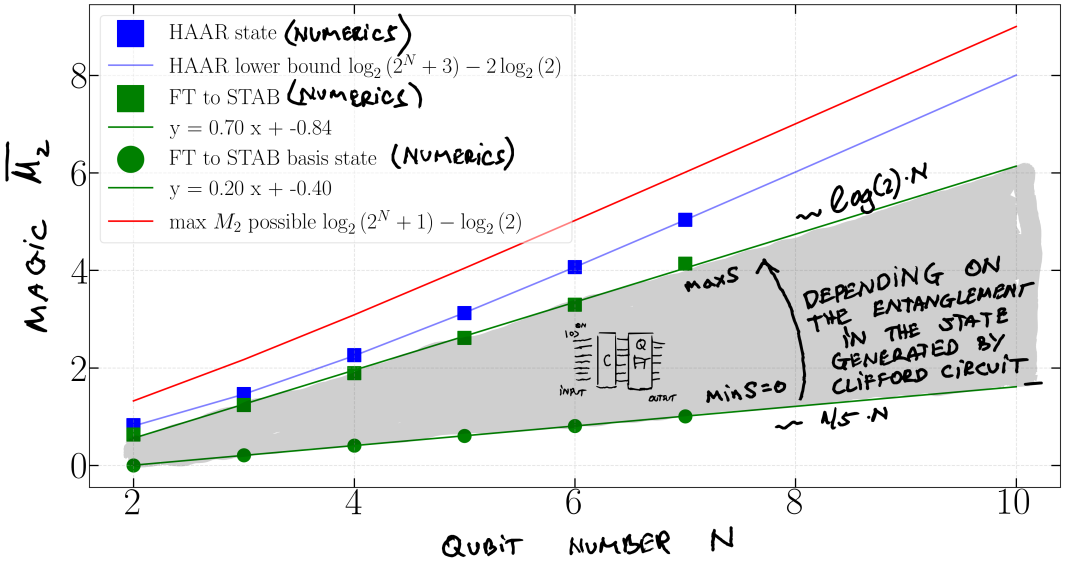


Figure 7: Numerical results

4 Numerical results

We evaluate non-stabilizing magic according to definition

$$\bar{M}_2(U^{\mathrm{FT}}) = \mathbb{E}_C \left[M_2(U^{\mathrm{FT}} C|0\rangle^{\otimes N}) \right], \tag{4.1}$$

where have used the definition for the SRE as

$$M_2(|\psi\rangle) = -\log_2 \left[\frac{1}{2^N} \sum_{P \in \mathbb{P}(N)} \mathrm{Tr}^4 \left[P|\psi\rangle\langle\psi| \right] \right]. \tag{4.2}$$

In Eq. (4.1) the initial state $|0\rangle^{\otimes N}$ is pure and has no magic. The state evolved under random Clifford circuit of depth N_L under $C = \prod_{i=1}^{N_L} U_k$, where U_k contains $N-1$ Clifford gates such as Hadamard, Phase S and CNOT gate between any pair of spins/qubits. The parameter choices are number of Clifford layers $N_L = 50N^2$. The results are present in Fig. 7.

A Nested commutators

In terms of layers we can split the commutator and the anti-commutators as

$$[U_3U_2U_1, P] = U_3U_2[U_1, P] + U_3[U_2, P]U_1 + [U_3, P]U_2U_1, \tag{A.1}$$

$$\{U_3U_2U_1, P\} = U_3U_2\{U_1, P\} - U_3[U_2, P]U_1 - [U_3, P]U_2U_1. \tag{A.2}$$

In full generality we obtain for the commutators

$$[U_1U_2 \dots U_n, P] = \sum_{j=1}^{n} \left(U_1U_2 \cdots U_{j-1}\right) \left[U_j, P\right] \left(U_{j+1}U_{j+2} \cdots U_n\right)$$

$$\{U_1U_2 \cdots U_n, P\} = U_1U_2 \dots U_{n-1}\{U_n, P\} - \sum_{j=1}^{n-1} \left(U_1U_2 \cdots U_{j-1}\right) \left[U_j, P\right] \left(U_{j+1}U_{j+2} \cdots U_{n-1}\right) \tag{A.3}$$

B Not necessary calculation

NOTE A: Lets first evaluate the application of the first layer. In particular,

$$\sum_{P(2)} ([U_1, P] \otimes \{U_1, P\} \otimes \{U_1, P\} \otimes \{U_1, P\}) \tag{B.1}$$

Where $P = A \otimes B$ while we use notation $A, B \in \{I, X, Y, Z\}$). The commutator simplifies as

$$\begin{aligned}
[U_1, P] &= [H \otimes I, A \otimes B] \\
&= \frac{1}{2}([H, A] \otimes \{I, B\} + \cancelto{0}{\{H, A\} \otimes [I, B]}), \\
&= \frac{1}{2}([H, A] \otimes \{I, B\}).
\end{aligned} \tag{B.2}$$

where in the second term the commutator $[I, B]$ is always zero. While the anticommutator reads after the same simplification

$$\{U_1, P\} = \frac{1}{2}(\{H, A\} \otimes \{I, B\}). \tag{B.3}$$

Plugging these relations in Eq. (B.1) we obtain

$$\frac{1}{16} \sum_{A,B \in \{I,X,Y,Z\}} \Big([H, A] \otimes \{I, B\}\Big) \otimes \Big(\{H, A\} \otimes \{I, B\}\Big) \otimes \Big(\{H, A\} \otimes \{I, B\}\Big) \otimes \Big(\{H, A\} \otimes \{I, B\}\Big). \tag{B.4}$$

Listing first 4 possibilities

$$\begin{aligned}
&\Big([H, I] \otimes \{I, I\}\Big) \otimes \Big(\{H, I\} \otimes \{I, I\}\Big) \otimes \Big(\{H, I\} \otimes \{I, I\}\Big) \otimes \Big(\{H, I\} \otimes \{I, I\}\Big), \\
&\Big([H, I] \otimes \{I, X\}\Big) \otimes \Big(\{H, I\} \otimes \{I, X\}\Big) \otimes \Big(\{H, I\} \otimes \{I, X\}\Big) \otimes \Big(\{H, I\} \otimes \{I, X\}\Big), \\
&\Big([H, I] \otimes \{I, Y\}\Big) \otimes \Big(\{H, I\} \otimes \{I, Y\}\Big) \otimes \Big(\{H, I\} \otimes \{I, Y\}\Big) \otimes \Big(\{H, I\} \otimes \{I, Y\}\Big), \\
&\Big([H, I] \otimes \{I, Z\}\Big) \otimes \Big(\{H, I\} \otimes \{I, Z\}\Big) \otimes \Big(\{H, I\} \otimes \{I, Z\}\Big) \otimes \Big(\{H, I\} \otimes \{I, Z\}\Big),
\end{aligned} \tag{B.5}$$

are all zero because $[H, I] = 0$. The next 4 read

$$\begin{aligned}
&\Big([H, X] \otimes \{I, I\}\Big) \otimes \Big(\{H, X\} \otimes \{I, I\}\Big) \otimes \Big(\{H, X\} \otimes \{I, I\}\Big) \otimes \Big(\{H, X\} \otimes \{I, I\}\Big), \\
&\Big([H, X] \otimes \{I, X\}\Big) \otimes \Big(\{H, X\} \otimes \{I, X\}\Big) \otimes \Big(\{H, X\} \otimes \{I, X\}\Big) \otimes \Big(\{H, X\} \otimes \{I, X\}\Big), \\
&\Big([H, X] \otimes \{I, Y\}\Big) \otimes \Big(\{H, X\} \otimes \{I, Y\}\Big) \otimes \Big(\{H, X\} \otimes \{I, Y\}\Big) \otimes \Big(\{H, X\} \otimes \{I, Y\}\Big), \\
&\Big([H, X] \otimes \{I, Z\}\Big) \otimes \Big(\{H, X\} \otimes \{I, Z\}\Big) \otimes \Big(\{H, X\} \otimes \{I, Z\}\Big) \otimes \Big(\{H, X\} \otimes \{I, Z\}\Big).
\end{aligned} \tag{B.6}$$

These 4 provide:

$$\begin{aligned}
&128i\sqrt{2}\Big(Y \otimes I \otimes I \otimes I \otimes I \otimes I \otimes I \otimes I\Big) \\
&128i\sqrt{2}\Big(Y \otimes X \otimes I \otimes X \otimes I \otimes X \otimes I \otimes X\Big) \\
&128i\sqrt{2}\Big(Y \otimes Y \otimes I \otimes Y \otimes I \otimes Y \otimes I \otimes Y\Big) \\
&128i\sqrt{2}\Big(Y \otimes Z \otimes I \otimes Z \otimes I \otimes Z \otimes I \otimes Z\Big)
\end{aligned} \tag{B.7}$$

The next 4 are

$$\begin{aligned}
&\Big([H,Y]\otimes\{I,I\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,I\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,I\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,I\}\Big)\\
&\Big([H,Y]\otimes\{I,X\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,X\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,X\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,X\}\Big)\\
&\Big([H,Y]\otimes\{I,Y\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,Y\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,Y\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,Y\}\Big)\\
&\Big([H,Y]\otimes\{I,Z\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,Z\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,Z\}\Big)\otimes\Big(\{H,Y\}\otimes\{I,Z\}\Big)
\end{aligned}\tag{B.8}$$

and are zero as $\{H,Y\}=0$. The final 4 are

$$\begin{aligned}
&\Big([H,Z]\otimes\{I,I\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,I\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,I\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,I\}\Big)\\
&\Big([H,Z]\otimes\{I,X\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,X\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,X\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,X\}\Big)\\
&\Big([H,Z]\otimes\{I,Y\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,Y\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,Y\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,Y\}\Big)\\
&\Big([H,Z]\otimes\{I,Z\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,Z\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,Z\}\Big)\otimes\Big(\{H,Z\}\otimes\{I,Z\}\Big)
\end{aligned}\tag{B.9}$$

and read

$$\begin{aligned}
&-128i\sqrt{2}\Big(Y\otimes I\otimes I\otimes I\otimes I\otimes I\otimes I\otimes I\Big)\\
&-128i\sqrt{2}\Big(Y\otimes X\otimes I\otimes X\otimes I\otimes X\otimes I\otimes X\Big)\\
&-128i\sqrt{2}\Big(Y\otimes Y\otimes I\otimes Y\otimes I\otimes Y\otimes I\otimes Y\Big)\\
&-128i\sqrt{2}\Big(Y\otimes Z\otimes I\otimes Z\otimes I\otimes Z\otimes I\otimes Z\Big)
\end{aligned}\tag{B.10}$$

All together we obtain

$$\sum_{P(2)}\left([U_1,P]\otimes\{U_1,P\}\otimes\{U_1,P\}\otimes\{U_1,P\}\right)=0\tag{B.11}$$

and other 7 terms vanish as well in the similar manner because its just permutation of the terms above. Similar calculation for the U_3 also gives zero for commutator.

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